

HITESH GOYAL

+65 86226962 | hiteshgoyal.business@gmail.com | [linkedin.com/in/hitesh-goyal-5732bb208](https://www.linkedin.com/in/hitesh-goyal-5732bb208) | github.com/indianeagle4599

ABOUT

Performance-driven AI Engineer and NTU graduate with a strong foundation in identifying and resolving system bottlenecks, to transform complex AI models into stable, optimised solutions. Skilled in the end-to-end model lifecycle—from training and fine-tuning to re-engineering pipelines for maximum resource efficiency to bridge the gap between research prototypes and functional tools by developing modular, clean code and managing containerized environments via Docker and AWS.

EDUCATION

Nanyang Technological University <i>Master of Science in Artificial Intelligence (CGPA: 3.89)</i>	Singapore Jan. 2026
Vellore Institute of Technology <i>B.Tech in Computer Science (Specialization in AI/ML, CGPA: 9.06)</i>	Chennai, India Jul. 2023

WORK EXPERIENCE

Graduate AI Engineering Intern (Summation Fellow) <i>Hummingbird Bioscience (SGInnovate Deep-Tech Programme)</i>	Aug. 2025 – Jan. 2026 Singapore
- Achieved a 50× runtime reduction in cancer-target discovery by implementing PyTorch-GPU vectorization and re-engineering data augmentation logic. Tripled feature deployment throughput on HPC clusters by architecting a config-driven SLURM orchestration tool for interactive GPU sessions. Orchestrated Generative AI deployment (BoltzGen) on AWS EC2 using DL AMIs for stable protein binder inference. 90% reduction in deck-creation time (2 hrs to 10 mins) through a custom interactive visualization system with persistent dashboards.	
AI Engineer Full Time (Computer Vision) <i>Tata Elxsi</i>	Jul. 2023 – Jun. 2024 Bangalore, India
Secured a new client by engineering a real-time CV/clustering anomaly detection pipeline for pollution monitoring. - Achieved 90% recall and 80% precision in content moderation using a DP-inspired temporal smoothing algorithm; parallelized it with Celery for distributed queuing. - Enabled lag-free ad placement at 30+ FPS with stream-agnostic inference logic ; boosted object tracking throughput by 20× via batching. - Achieved 78.1% mAP on industrial defect segmentation; outperformed task-specific baselines on unseen data using a curated 30k-image dataset.	
AI Intern (Research & Development) <i>Samsung R&D</i>	Nov. 2021 – Oct. 2022 Bangalore, India
Led R&D for 5 peers on frame-rate conversion; exceeded speed and quality KPIs by 1.5×. 27.9 PSNR achieved on 448x256 video sequences; managed research milestones and senior technical presentations.	

PROJECTS

LazyResume: OCR & LLM Document Parsing <i>Vision-LLMs, FastAPI</i>	May 2025
- Engineered a pipeline using Vision-based OCR and LLMs; designed to bridge the recruiter-candidate gap by automating unstructured data parsing into structured JSON. - Improved parseability of complex layouts by 20% via custom evaluation hooks for resume-to-JD alignment.	
Fuzzy Hybrid Model for Stock Trading <i>Python, Scikit-Learn, NumPy</i>	Sep. 2024
- Reduced feature dimensionality by 5× while improving model performance by approximately 20%. - Optimized data pipelines to improve loading and training speed by 10×.	

TECHNICAL SKILLS

Frameworks: Python-PyTorch, TensorFlow, OpenCV, Scikit-Learn, NumPy, Pandas, LangChain, FastAPI
Infrastructure & MLOps: AWS (EC2, DL AMI), Docker, SLURM, Celery, Distributed Training, Quantization
Focus Areas: Deep Learning, Computer Vision, Generative AI, Video Processing, Object Detection